Experiment E: QKVO LoRA on All Four Attention Projections

Status: Completed; evaluated on 12 October 2026. Task-completion candidate, not the balanced reference.

Experiment index

Question and Controlled Change

Does adapting key and output projections alongside query and value projections improve task completion and balanced routing under A's original two-epoch recipe?

Setting	A: reference	E: candidate
Target modules	q_proj, v_proj	q_proj, k_proj, v_proj, o_proj
Rank / alpha	8 / 16	8 / 16
Learning rate	0.0002	0.0002
Epochs	2	2
Initialization	Fresh pretrained Qwen + fresh LoRA	Fresh pretrained Qwen + fresh LoRA

Only the target-module list changes, apart from run identity and output paths. E is not the rank-16 recipe or the lower-learning-rate recipe. Broader coverage also increases trainable adapter parameters: this is not a constant-parameter coverage comparison. Its result cannot attribute a gain solely to module placement.

Fixed Recipe

- Base model: Qwen/Qwen2.5-1.5B-Instruct; pretrained weights are not reinitialized.
- Frozen data: xlam-baymax-target-compatible-v14-act200-anchored-val200.
- Revision: 329fab4d8f1604a4085d3ec50e05881e7c069717bb1428db7e592743bb07f03c.
- Prompt SHA256: a31446de6c8b67f1fc8ba8fc09ed8e2d7425ba832334dc8fd6fe2d55ef978468.
- Raw training / validation caps: 1,000 / 200.
- Actual command overrides: context 1,024, train/eval batch one, accumulation 16.
- Seed 1, dropout 0.05, cosine scheduling with 3% warmup, existing PEFT initialization.
- The same assistant-label masking, target-contract checks and strict inference parser.

The JSON retains A's declared context and batch settings. The command applies the same actual low-VRAM overrides as A. A retained 974 training and 195 validation rows. E's final usable-row counts are not embedded in its evaluation JSON; the reference counts are not new measurements of E.

Training Command

Run from the repository root in PowerShell:

```
$config =
"configs\sft\qwen2_5_1_5b_lora_target_compatible_v14_act200_anchored_val200_epochs2_qkvo.json"
.venv\Scripts\python scripts\train_sft_lora.py `
  --config $config `
  --local-low-vram `
  --max-train-rows 1000 `
  --max-eval-rows 200 `
  --max-sequence-length 1024
```

The separate output directory ends in `epochs2_qkvo`. No previous model is overwritten. Additional adapted modules may increase memory pressure. Do not silently reduce sequence length or change quantization to finish this run; record any such change as a different experiment.

Results

Both runs use the established combined v1 + v2 development suite: 150 cases with 81 ACT, 40 CLARIFY and 29 REFUSE expectations. The owner command uses the same prompt, greedy decoding, 256 new tokens and the existing strict parser.

Metric	A: query/value only	E: query/key/value/output
Full task successes	44/150 (0.293)	52/150 (0.347)
Routing macro-F1	0.514	0.437
Invalid outputs	2/150 (0.013)	14/150 (0.093)
Tool-call accuracy	0.600	0.527
Argument accuracy	0.771	0.748
Clarification wording accuracy	0.075	0.113
Refusal wording accuracy	0.172	0.086
Decision accuracy among known decisions	0.568	0.581
Decision coverage	0.987	0.907
Unavailable-tool hallucination rate	0.020	0.053
Average latency (ms)	1828	3794
Reported API cost (USD)	0.0000	0.0000

Ratios are rounded to three decimals. Wording checks are separate from routing. Decision accuracy excludes unknown decisions; read it alongside coverage. Zero reported API cost excludes local hardware, electricity and training cost.

Routing Counts

Measurement	A	E
Correct ACT decisions	57/81	63/81
Correct CLARIFY decisions	11/40	10/40
Correct REFUSE decisions	16/29	6/29
Premature ACT on boundary cases	31/69	40/69
Invalid/unknown decisions	2/150	14/150

Premature ACT here is the count derived from the CLARIFY/REFUSE matrix rows, not necessarily the scorer's failure-reason count.

A: Reference Decision Matrix

Rows are expected decisions; columns are recorded predictions. The diagonal measures routing correctness, not full task success.

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	57	21	3	0
CLARIFY	20	11	9	0
REFUSE	11	0	16	2

E: QKVO Decision Matrix

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	63	12	2	4
CLARIFY	27	10	2	1
REFUSE	13	1	6	9

Invalid-Output Inspection

Observed issue	Expected cases	Responses
refuse_request put reason outside arguments	REFUSE	6
request_clarification had empty arguments	CLARIFY 1, REFUSE 2	3
Productive calls omitted or separated names from arguments	ACT	4
Other malformed output	REFUSE	1

Six of the nine invalid REFUSE-case outputs selected the refusal meta-tool name, but supplied reason at the call's top level instead of inside arguments:

```
{"name": "refuse_request", "arguments": {}, "reason": "Unsupported operation."}
```

The contract requires:

```
{"name": "refuse_request", "arguments": {"reason": "Unsupported operation."}}
```

These are generic illustrations of the nesting defect, not repaired benchmark responses. Selecting refuse_request does not establish that its explanation meets the scenario's wording requirements. No rejected response received success credit.

Interpretation and Decision

E completed eight more tasks than A, the largest task-success gain among the A/B/C/E ablations. However, macro-F1 fell, invalid outputs rose, and latency was about 2.1 times A's measured latency. E is not a better balanced replacement.

Keep A as the balanced reference and E as a task-completion candidate. A substantial part of E's observed refusal loss involves invalid expression of an attempted refusal. This supports investigating output-contract learning separately; it does not erase the other incorrect decisions or establish a hypothetical repaired score.

The next proposed test is an independently authored output-contract learning experiment with the policy class quantities and other training settings held fixed. No data adjustment or new run has been made by this documentation task.

Provenance and Limits

- E evaluation JSON: recorded 2026-10-12T19:19:35.198791+00:00.
- A evaluation JSON: recorded 2026-10-06T15:29:45.372202+00:00.
- E canonical config SHA256: ee2c1b9efdec057796d0c0a9d16b20f75763151fcc34e6fe95ec004bc3bf00e4.

Metrics and matrices were checked with the existing comparison helper. Raw generations were inspected read-only. Original scores, results and parser rules remain unchanged. These are single-seed development observations, not an untouched final test. Adding modules also increases adapter size, so this does not isolate placement at equal parameter count. Full training/generation metadata and exact base-model revision are not embedded in the result JSON.

Supporting Files

- E config
- A: selected reference

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.